Supplemental Table 1: Data types commonly used in genomic approaches to species delimitation. Data types can be categorized into two general classes: those in which a subset of the genome is sequenced (reduced representation approaches) and those in which the full genome is sequenced (whole genome approaches).

General Class	Specific Type	Description	Resources required
Reduced representation approaches	Restriction-aided approaches	Genomic DNA is digested with restriction enzymes; digested DNA is size-selected and sequenced	None
	Target capture	Baits are used to capture a subset of the genome; baits can be designed to any subset of loci (e.g., exons, single-copy genes, restriction-aided loci)	Baits – which can be synthesized based on sequence or generated anonymously
	Transcriptome sequencing	mRNA is extracted and sequenced	RNA preserved tissues
Whole genome approaches	Whole genome sequencing	*de novo* sequencing and assembly of individuals of interest	None, but if genomes are large, high-memory & high-storage computational power
	Whole genome re-sequencing	Sequencing of the full genome for individuals for which a relevant reference genome already exists	Reference genome
	Genome skimming	Sequencing of the full genome for individuals at very low coverage (<1x)	None but works better with reference genome

Supplemental Table 2: Analytical methods that can be used for species delimitation. Approaches are grouped according to the two primary steps of species delimitation: (1) identifying putative species level units and (2) evaluating the robustness of these putative species. This list reflects the common methods identified in our review (see "Genomic Species Delimitation in Practice: A Review of Genomic Species Delimitation") and other less commonly used methods of interest.

Step	General Class	Method	Program	Description	Citation
Step 1:Identifying putative species-level units	Phylogenetic Methods	Manual Identification of Putative Species (monophyletic clades)	ASTRAL	infers species trees from a set of gene trees; can be used to infer individual level phylogenies and thus identify putative species-level clades	10.1186/s12859-018-2129-y
			IQTree	uses maximum likelihood to infer phylogenies from (phylogenomic) alignments; can be used to infer individual level phylogenies and thus identify putative species-level clades	10.1093/molbev/msaa015
			RAxML & ExaML	uses maximum likelihood to infer phylogenies from (phylogenomic) alignments; can be used to infer individual level phylogenies and thus identify putative species-level clades	10.1093/bioinformatics/btu033
			SNAPP	infers species trees from biallelic variants; can be used to infer individual level phylogenies and thus identify putative species-level clades	10.1093/molbev/mss086
			SVDquartets	infers species trees from variants; can be used to infer individual level phylogenies and thus identify putative species-level clades	10.1093/bioinformatics/btu530
		Automatic Identification of Putative Species (monophyletic clades)	gmyc	identifies the threshold between within-species and between-species coalescent events in an ultrametric, individual level tree to delimit species-level clades; while designed for phylogenies inferred from single loci, has been applied to phylogenomic trees	10.1093/sysbio/syt033

			PTP & bPTP & mPTP	identifies the threshold between within-species and between-species branch lengths in a non-ultrametric, individual level tree to delimit species-level clades; while designed for phylogenies inferred from single loci, has been applied to phylogenomic trees	10.1093/bioinformatics/btt499
	Genotypic Methods	Descriptive Genetic Clustering	adegenet (pca & dapc)	ordinates variant data to identify genetic similarities between individuals, which can then be used to identify putative genetic clusters	10.1093/bioinformatics/btn129
			pcANGSD	ordinates low coverage variant data to identify genetic similarities between individuals, which can then be used to identify putative genetic clusters	10.1534/genetics.118.301336
			PLINK	ordinates variant data to identify genetic similarities between individuals, which can then be used to identify putative genetic clusters	10.1093/bioinformatics/bts606
		Model-Based Genetic Clustering	ADMIXTURE	identifies genetic clusters by minimizing linkage & Hardy-Weinberg disequilibrium across a set of sampled individuals; genetic clusters can be treated as putative species	10.1101/gr.094052.109
			fastSTRUCTURE	identifies genetic clusters by minimizing linkage & Hardy-Weinberg disequilibrium across a set of sampled individuals; genetic clusters can be treated as putative species	10.1534/genetics.114.164350
			fineSTRUCTURE	uses linkage disequilibrium among variants to identify more fine-scale structure; identified genetic clusters can be treated as putative species	10.1371/journal.pgen.1002453
			sNMF	identifies genetic clusters based on sparse non-negative matrix factorization across a set of sampled individuals; genetic clusters can be treated as putative species	10.1534/genetics.113.160572

			structure	identifies genetic clusters by minimizing linkage & Hardy-Weinberg disequilibrium across a set of sampled individuals; genetic clusters can be treated as putative species	10.1093/genetics/155.2.945
		Gaussian Mixture Model Clustering	prabclus	transforms genotype data into distances based on allele frequencies and sharing; clusters are then identified via Gaussian mixture modeling and can be treated as putative species	10.1093/sysbio/syq039
Step 2: Evaluating robustness of putative species	Non-Tree Based Methods	Threshold methods: Calculating genetic divergence	Dashing	estimates (very) quickly nucleotide dissimilarity between full genomes; levels of dissimilarity can then be used to identify discontinuities that define a species-level threshold	10.1186/s13059-019-1875-0
			FastANI	estimates (very) quickly nucleotide dissimilarity between full genomes; levels of dissimilarity can then be used to identify discontinuities that define a species-level threshold	10.1038/s41467-018-07641-9
			gsi	provides a proxy for the extent of genealogical sorting, or the extent to which a given group has achieved monophyly; can be used to estimate genetic divergence among groups to assess species status	10.1111/j.1558-5646.2008.00442.x
			MASH	estimates (very) quickly nucleotide dissimilarity between full genomes; levels of dissimilarity can then be used to identify discontinuities that define a species-level threshold	10.1186/s13059-016-0997-x
		Machine Learning	CLADES	uses machine learning approaches on genomic datasets to determine if empirical data better matches a one species or two species model; machine learning model is trained on simulated data	10.1111/1755-0998.12887
			kohonen	uses machine learning approaches on self-organized maps across different metrics (including genetics, morphology) to identify species-level units	10.1016/j.ympev.2023.107939

	Tree-Based Methods	Multispecies coalescent approaches	BFD*	estimates the species tree from biallelic loci and simultaneously identifies the species delimitation model that best fits the data	10.1093/sysbio/syu018
			BPP	a Bayesian approach that uses the multispecies coalescent to estimate the posterior probability of a species	10.1073/pnas.0913022107
			gdi / hhsd	uses the population sizes and divergence times estimated from the multispecies coalescent model (with or without migration or introgression) to identify the probability that two populations are distinct species	10.1093/sysbio/syw117; 10.1093/sysbio/syae050
			iBPP	an approach that builds on BPP to simultaneously analyze phenotypic and genetic data to estimate the posterior probability of a species	10.1111/evo.12582
			soda	uses a set of gene trees to identify the boundary between where coalescent events are random versus not; this boundary identifies species-level clades	10.1093/bioinformatics/btaa1010
			SPEEDEMON	extends the birth-death collapse model to allow speciation rate to vary through time; the resultant model identifies species-level units	10.1038/s42003-022-03723-z
			STACEY	infers the phylogeny for the user-defined clusters within a set of sampled individuals; clusters are collapsed resulting in both the species delimitation and species tree	10.1007/s00285-016-1034-0
			TR2	identifies species-level units by comparing topological variations across gene trees; distributions of topologies are skewed when individuals belong to different species	10.1093/sysbio/syw028

	Protracted speciation model	DELINEATE	builds on the protracted speciation model to estimate the likelihood that an incipient lineage graduates to a good species, which is then used to delimit species	10.1371/journal.pcbi.1008924
	Combined approaches to demographic modeling & species delimitation	delimitR	simulates data under a range of divergence models and then uses Random Forest approach to identify which model best fits the empirical data; divergence models differ both in demography and the number of species modeled	10.1111/evo.13919
		phrapl	estimates an approximate likelihood for an empirical dataset to a range of divergence models that vary both in the number of lineages modeled and their demographic history; by identifying the best-fitting model, we can infer the number of species and their demographic history	10.1093/sysbio/syw117
	Other	SpedeSTEM	infers the maximum likelihood tree for a species tree and then uses information theory to determine the most likely number of partitions (or putative new species) within described species	10.1111/j.1755-0998.2010.02947.x
Gene-Flow Based Methods	Inferring introgression during species divergence from topological and/or branch length variation	ADMIXTOOLS	applies f-statistics and D-statistics to infer likely admixture events between populations; inference of gene flow can be used to diagnose species limits	10.1534/genetics.112.145037
		Dsuite	calculates D-statistics to infer likely admixture events between populations; inference of gene flow can be used to diagnose species limits	10.1111/1755-0998.13265
		hyde	applies D-statistic type tests to gene trees inferred from phylogenomic loci to identify instances of introgression; inference of gene flow can be used to diagnose species limits	10.1093/sysbio/syy023

			QuIBL	uses the distribution of branch lengths to identify likely introgression events; inference of introgression can be used to diagnose species limits	10.1126/science.aaw2090
		Inferring levels of gene flow during species divergence through demographic modelling	BayesAss	estimates recent migration rates between populations; these migration estimates can be used to diagnose species limits	10.1093/genetics/163.3.1177; 10.1111/2041-210X.13252
			dadi	uses the site frequency spectrum between populations to infer demographic history; estimates of the extent of gene flow can be used to diagnose species limits	10.1371/journal.pgen.1000695
			fastsimcoal2	uses the site frequency spectrum between populations to infer demographic history; estimates of the extent of gene flow can be used to diagnose species limits	10.1093/bioinformatics/btab468
			G-PhoCS	uses multiple, unlinked loci to infer the demographic history of populations; estimates of the extent of gene flow can be used to diagnose species limits	10.1038/ng.937
			moments	uses the site frequency spectrum between populations to infer demographic history; estimates of the extent of gene flow can be used to diagnose species limits	10.1534/genetics.117.200493
			momi2	uses the site frequency spectrum between populations to infer demographic history; estimates of the extent of gene flow can be used to diagnose species limits	10.1080/01621459.2019.1635482
		Identifying introgression by inferring cline width at	bgc / bgchm	fits genomic clines across hybrid individuals; estimates of genomic introgression can be used to diagnose species limits	10.1101/2024.03.29.587395

		parapatric boundaries between putative species	HZAR	fits geographic clines across sampled transects; estimates of cline width can be used to diagnose species limits	10.1111/1755-0998.12209
		Identifying introgression by looking for admixed individuals (often at parapatric boundaries between putative species)	see programs listed under "Model-Based Genetic Clustering"	can identify individuals of likely admixed origins; the prevalence and distribution of these individuals can be used to diagnose species limits	10.1038/s41467-018-05257-7
		Tree-based inference of gene flow	MSC-I & MSC-M	extends the BPP approach to determine where - and to what extent - introgression and/or migration have occurred across a set of lineages	10.1073/pnas.2310708120
			PhyloNet	offers multiple methods to infer a phylogenetic network from unlinked loci or gene trees; evidence of introgression and hybridization across lineages can inform understanding of species boundaries	10.1093/sysbio/syy015
			PhyloNetworks	offers multiple methods to infer a phylogenetic network from unlinked loci or gene trees; evidence of introgression and hybridization across lineages can inform understanding of species boundaries	10.1093/molbev/msx235
			TreeMix	uses allele frequency patterns across a set of populations to infer the tree that best describes these populations and patterns of gene flow amongst them; inference of gene flow can be used to diagnose species limits	10.1038/npre.2012.6956.1